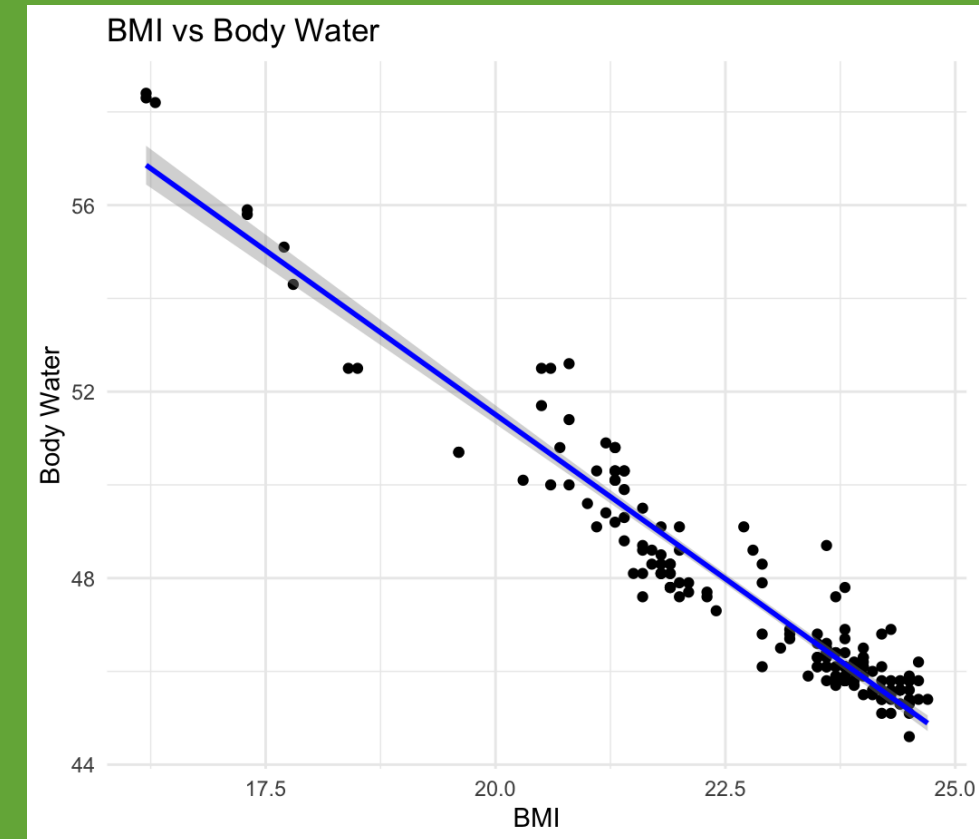
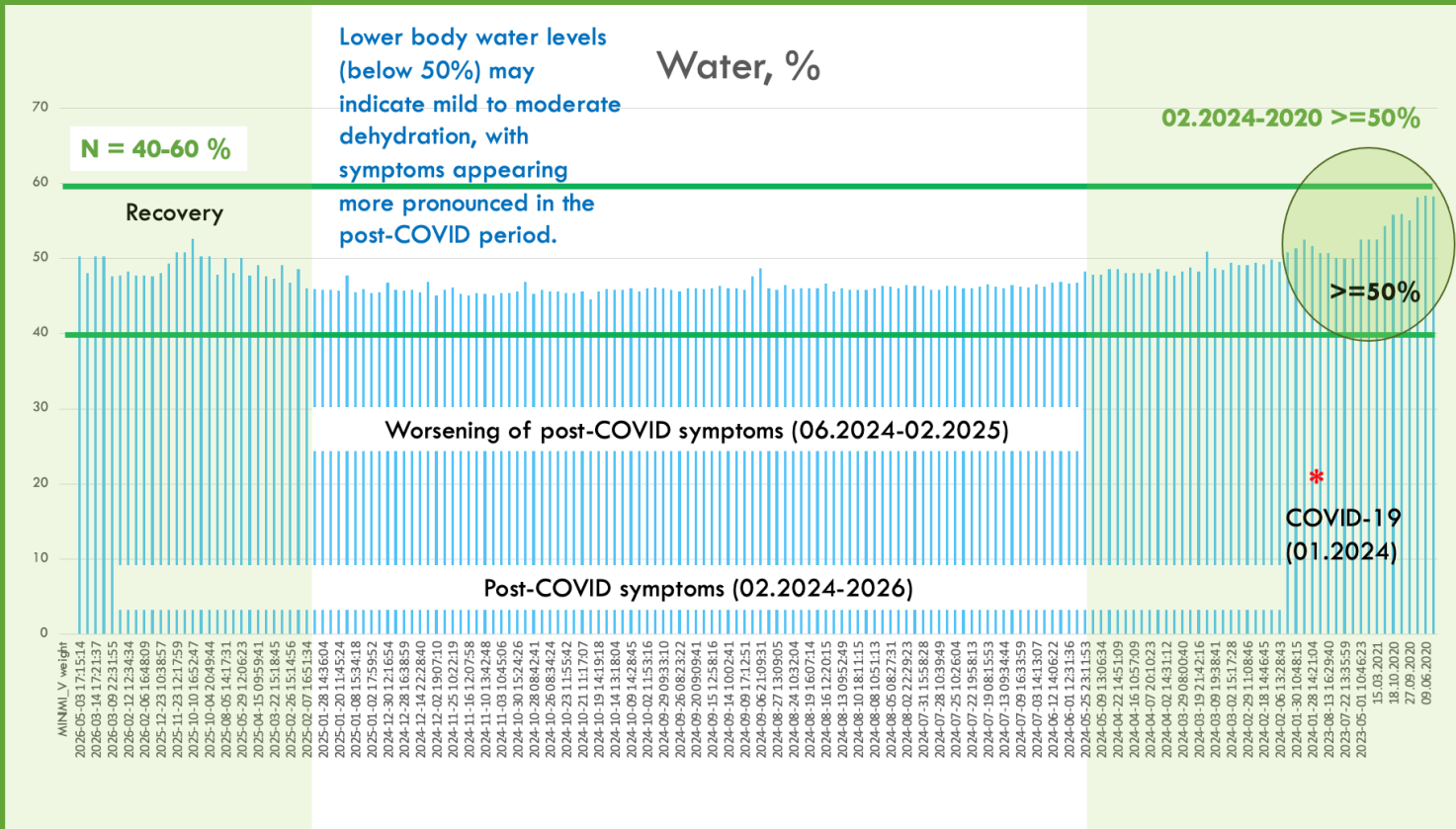




# BIA BODY COMPOSITION ANALYSIS DURING COVID-19

# BIA BODY COMPOSITION ANALYSIS DURING COVID-19

Exploratory analysis of body composition (BIA) data during the COVID-19 years using R.

Analysis of a real-world dataset ( $n > 100$ ) revealed a very strong negative correlation between **body fat** and **body water** ( $r = -0.99$ ,  $p < 0.001$ ), a very strong negative correlation between **BMI** and **body water** ( $r = -0.96$ ,  $p < 0.001$ ), and a very strong positive correlation between **BMI** and **body fat** ( $r = 0.96$ ,  $p < 0.001$ ).

The data represent repeated measurements from **a single individual** over time; therefore, the observed relationships likely reflect underlying physiological dependencies and measurement characteristics of BIA.



# TOOLS AND TECHNOLOGIES

Noerden Minimi (bioelectrical impedance scale)

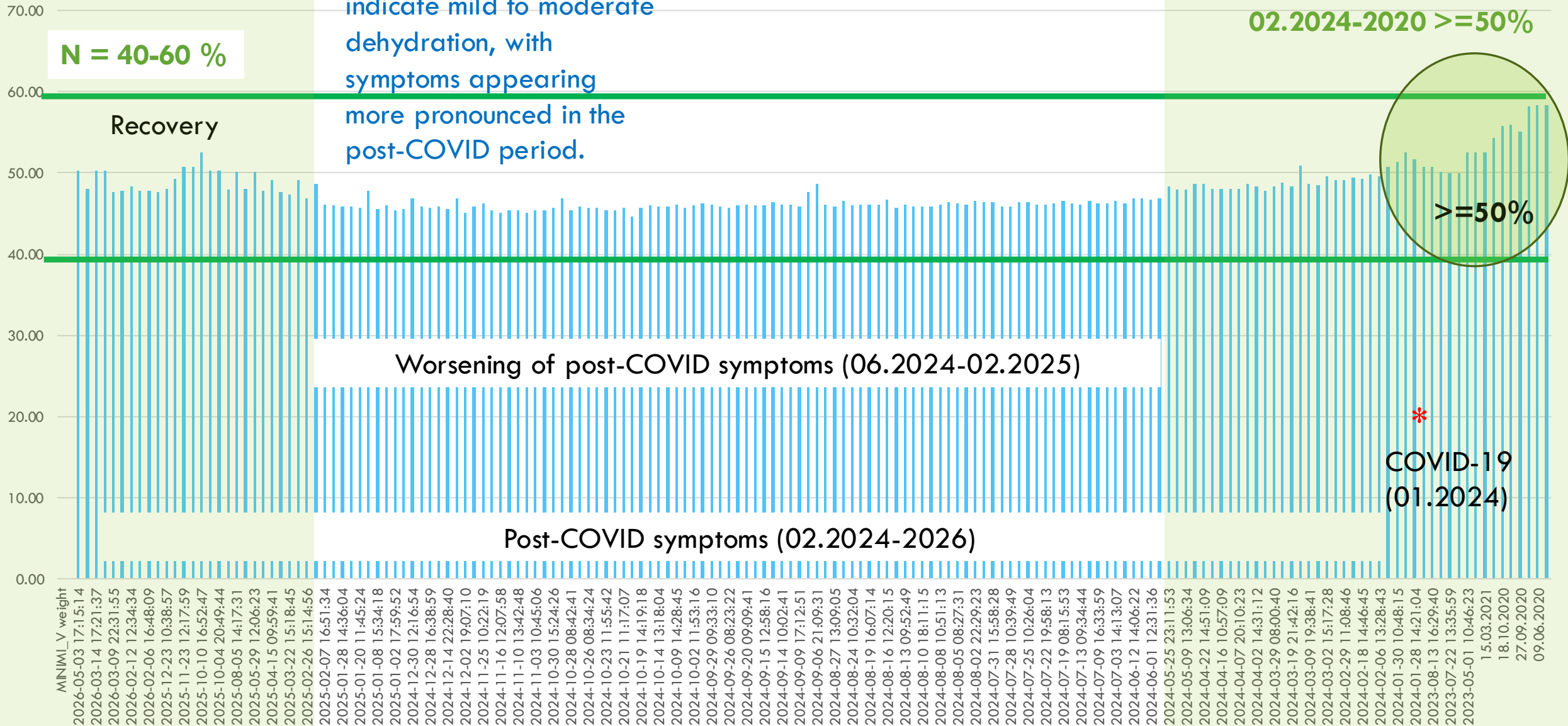
Microsoft Excel

Google Sheets

Microsoft PowerPoint

RStudio

# Water, %

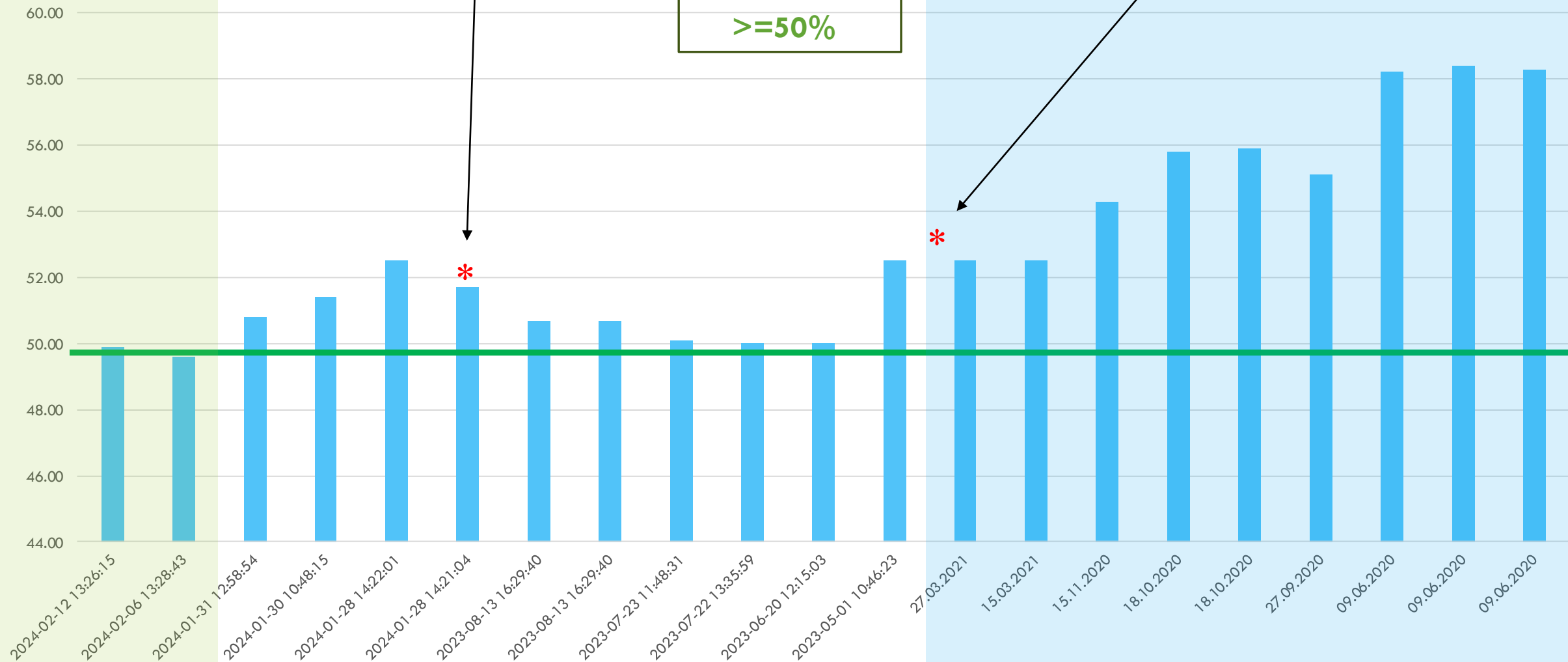


Post-COVID  
period  
(02.2024-2026)

COVID-19  
(01.2024)

Water, %  
≥50%

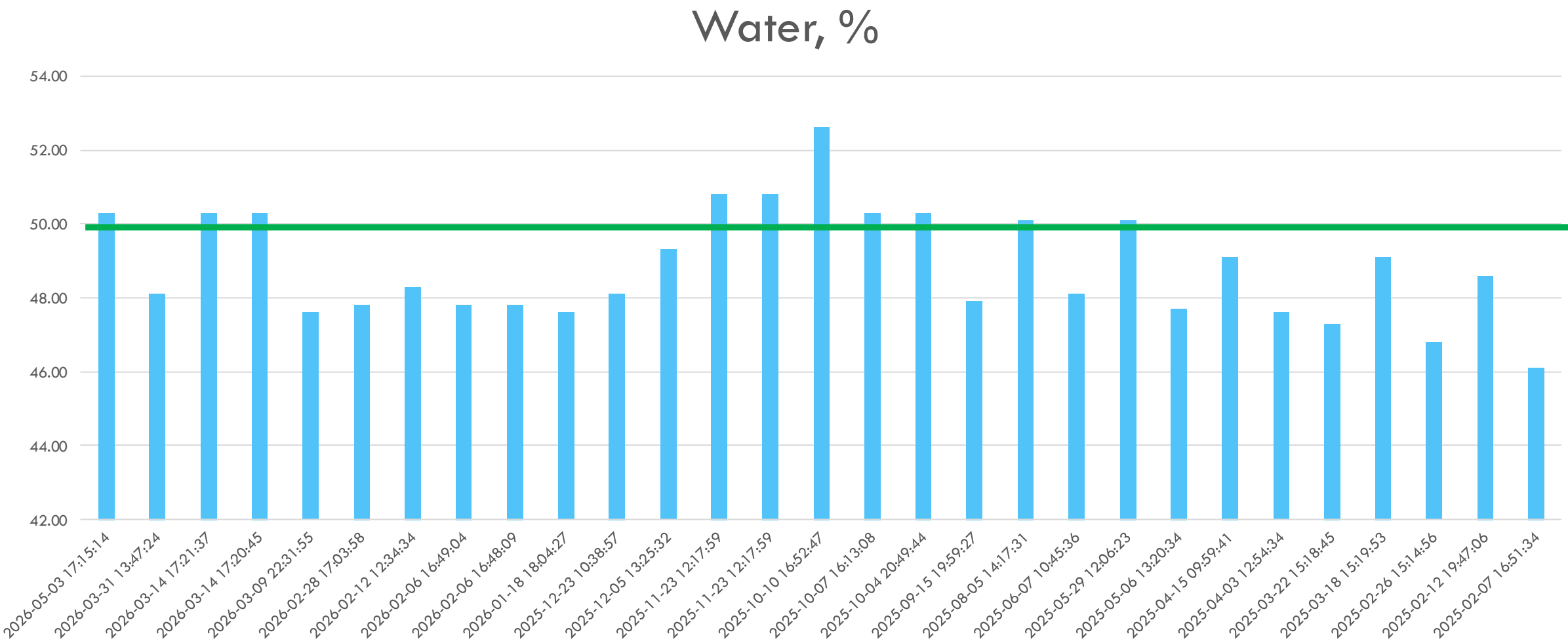
COVID-19  
(06.2021)



02.2024-05.2023

2021-2020 (old data)

Recovery after COVID-19 (2026-02.2025)

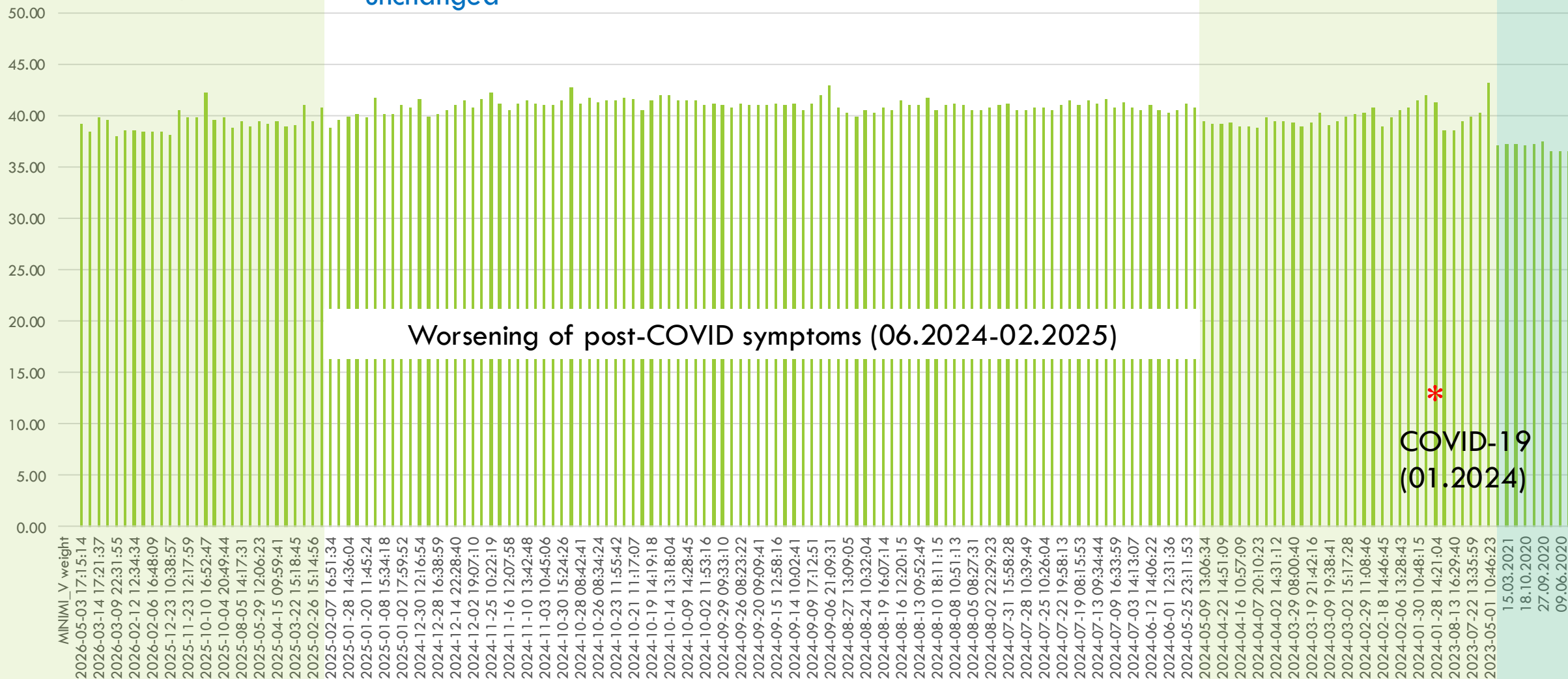


Post-COVID period  
(2026-02.2025),  
recovery

Muscle mass  
remained  
unchanged

Muscle Mass, kg

Low BMI, 2020-2021  
(old data)

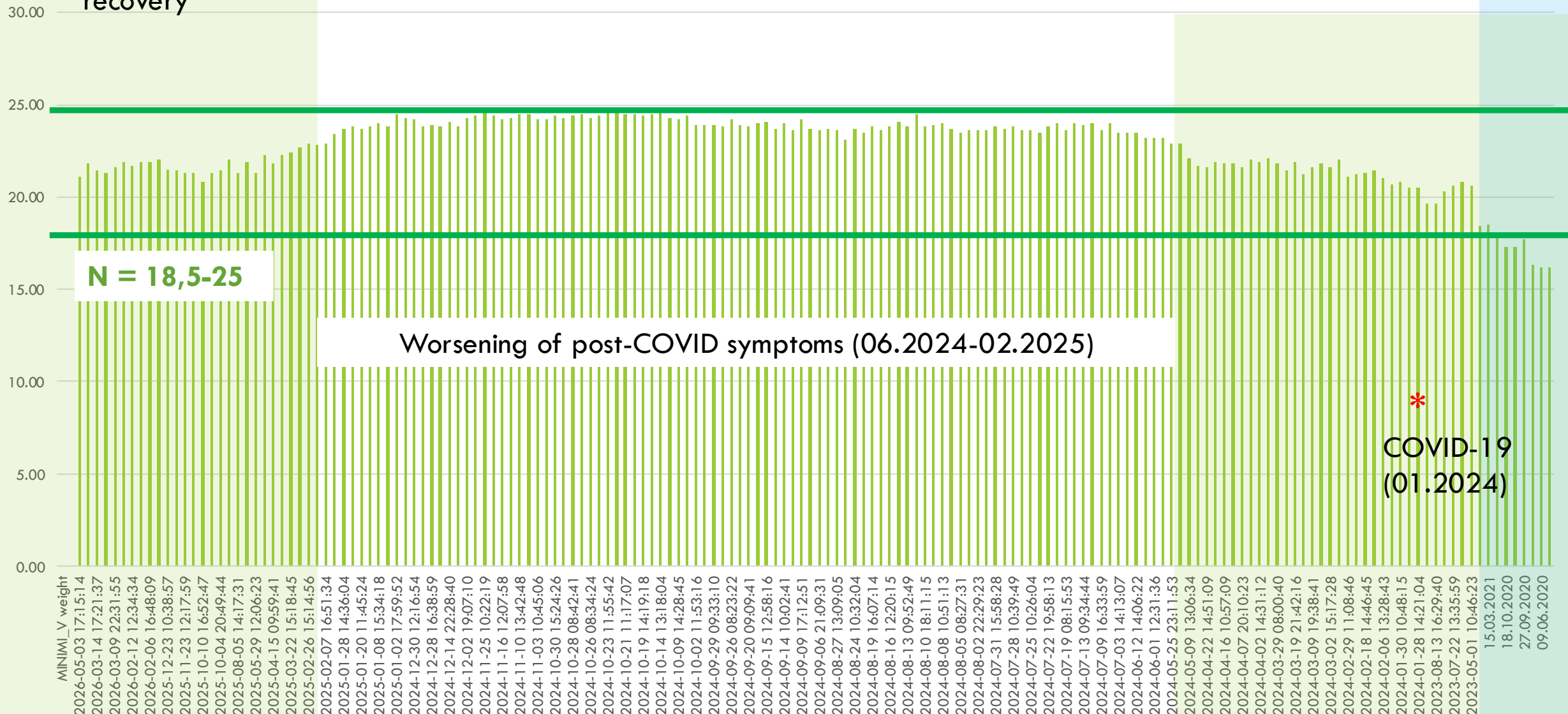


Post-COVID period  
(2026-02.2025),  
recovery

BMI increased due to an  
increase in **body fat**.

BMI

Low BMI, 2020-2021  
(old data)



Post-COVID period  
(2026-02.2025),

An increase in **body fat**  
was observed in the post-  
COVID period

Fat, %

Low BMI, 2020-2021  
(old data)

40.00 recovery

35.00

30.00

25.00

20.00

N = 21-33 %

Worsening of post-COVID symptoms (06.2024-02.2025)

15.00

10.00

5.00

0.00

MINIMI\_V weight

2026-05-03 17:15:14  
2026-03-14 17:21:37  
2026-03-09 22:31:55  
2026-02-12 12:34:34  
2026-02-06 16:48:09  
2025-12-23 10:38:57  
2025-11-23 12:17:59  
2025-10-10 16:52:47  
2025-10-04 20:49:44  
2025-08-05 14:17:31  
2025-05-29 12:06:23  
2025-04-15 09:59:41  
2025-03-22 15:18:45  
2025-02-26 15:14:56  
2025-02-07 16:51:34  
2025-01-28 14:36:04  
2025-01-20 11:45:24  
2025-01-08 15:34:18  
2025-01-02 17:59:52  
2024-12-30 12:16:54  
2024-12-28 16:38:59  
2024-12-14 22:28:40  
2024-12-02 19:07:10  
2024-11-25 10:22:19  
2024-11-16 12:07:58  
2024-11-10 13:42:48  
2024-11-03 10:45:06  
2024-10-30 15:24:26  
2024-10-28 08:42:41  
2024-10-26 08:34:24  
2024-10-23 11:55:42  
2024-10-21 11:17:07  
2024-10-19 14:19:18  
2024-10-14 13:18:04  
2024-10-09 14:28:45  
2024-10-02 11:53:16  
2024-09-29 09:33:10  
2024-09-26 08:23:22  
2024-09-20 09:09:41  
2024-09-15 12:58:16  
2024-09-14 10:02:41  
2024-09-09 17:12:51  
2024-09-06 21:09:31  
2024-08-27 13:09:05  
2024-08-24 10:32:04  
2024-08-19 16:07:14  
2024-08-16 12:20:15  
2024-08-13 09:52:49  
2024-08-10 18:11:15  
2024-08-08 10:51:13  
2024-08-05 08:27:31  
2024-08-02 22:29:23  
2024-07-31 15:58:28  
2024-07-28 10:39:49  
2024-07-25 10:26:04  
2024-07-22 19:58:13  
2024-07-19 08:15:53  
2024-07-13 09:34:44  
2024-07-09 16:33:59  
2024-07-03 14:13:07  
2024-06-12 14:06:22  
2024-06-01 12:31:36  
2024-05-25 23:11:53  
2024-05-09 13:06:34  
2024-04-22 14:51:09  
2024-04-16 10:57:09  
2024-04-07 20:10:23  
2024-04-02 14:31:12  
2024-03-29 08:00:40  
2024-03-19 21:42:16  
2024-03-09 19:38:41  
2024-03-02 15:17:28  
2024-02-29 11:08:46  
2024-02-18 14:46:45  
2024-02-06 13:28:43  
2024-01-30 10:48:15  
2024-01-28 14:21:04  
2023-08-13 16:29:40  
2023-07-22 13:35:59  
2023-05-01 10:46:23  
15.03.2021  
18.10.2020  
27.09.2020  
09.06.2020

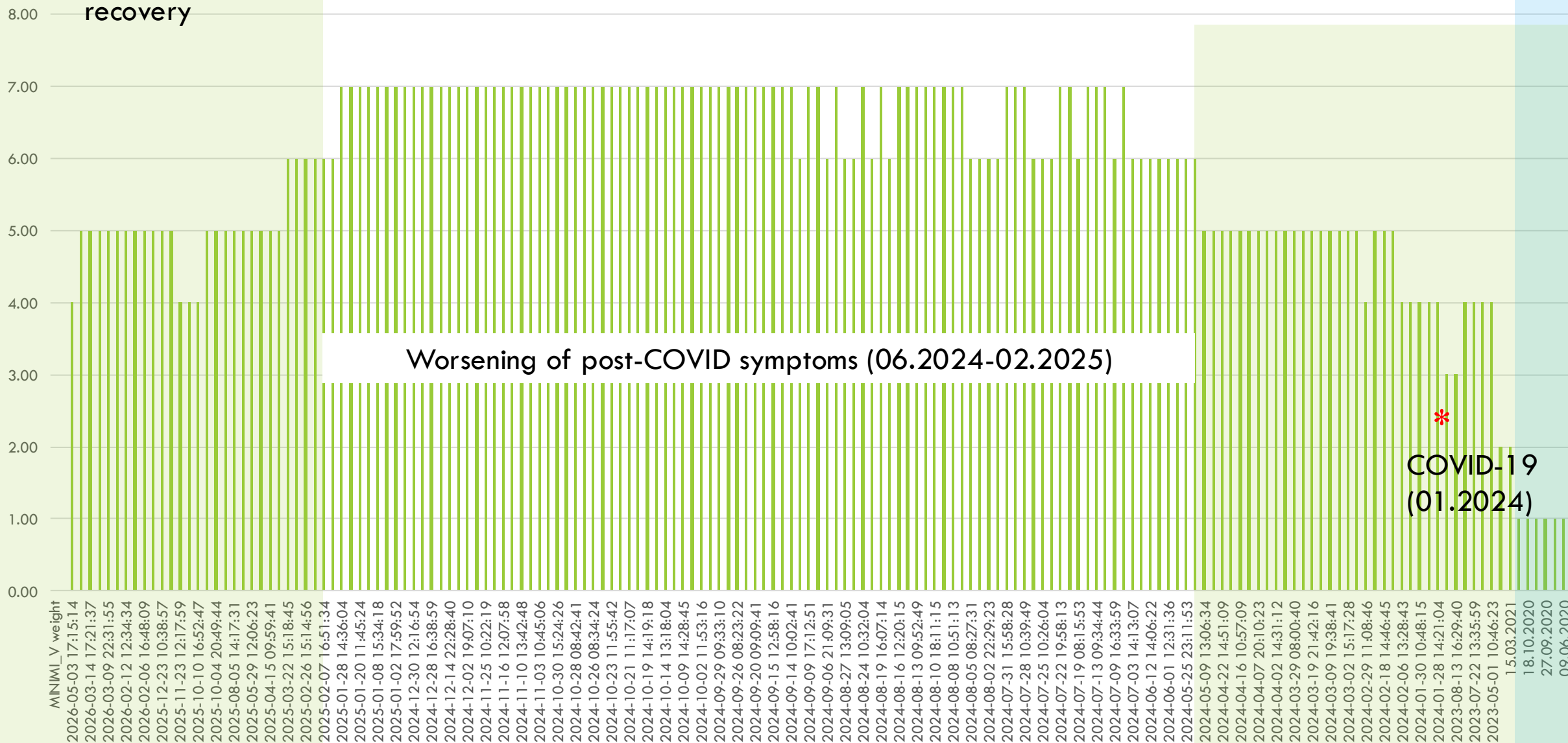
\*  
COVID-19  
(01.2024)

Post-COVID period  
(2026-02.2025),

N <=14

# Visceral Fat Index

Low BMI, 2020-2021  
(old data)

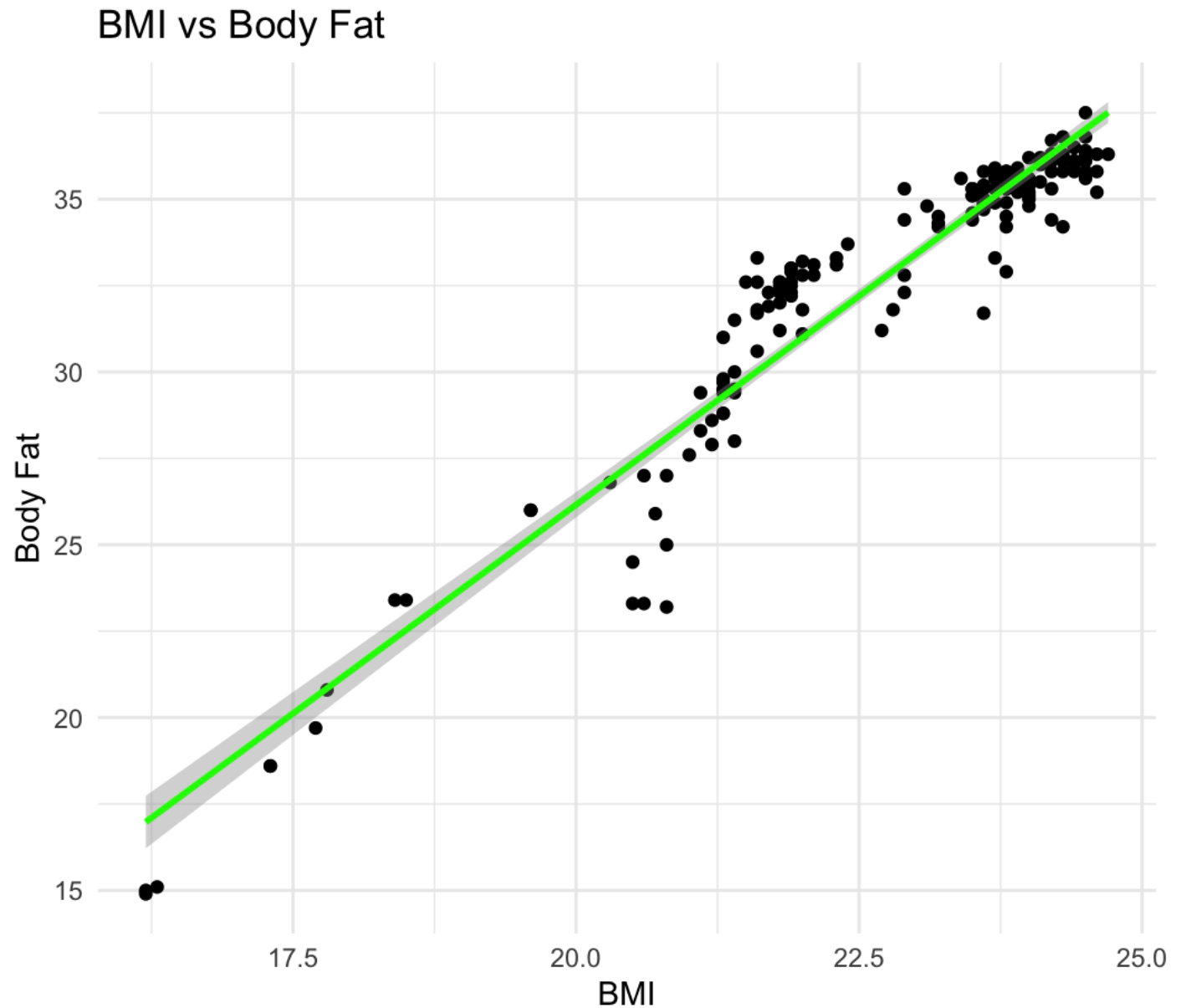


# RSTUDIO

```
> data <- read.csv("data_bia.csv")
> names(data)
[1] "bmi"    "fat"    "water"
> str(data)
'data.frame':  168 obs. of  3 variables:
 $ bmi   : num  21.1 21.8 21.4 21.3 21.6 21.9 21.7 21.9 21.9 22 ...
 $ fat   : num  29.4 32.6 29.4 29.4 33.3 32.9 32.3 33 33 33.2 ...
 $ water: num  50.3 48.1 50.3 50.3 47.6 47.8 48.3 47.8 47.8 47.6 ...
> cor(data$bmi, data$fat)
[1] 0.9565598
> cor(data$fat, data$water)
[1] -0.9888601
> cor(data$bmi, data$water)
[1] -0.9613819
```

```
library(ggplot2)

ggplot(data, aes(x = bmi, y = fat)) +
  geom_point() +
  geom_smooth(method = "lm",
    color = "green") +
  labs(title = "BMI vs Body Fat",
    x = "BMI",
    y = "Body Fat") +
  theme_minimal()
```



```
ggplot(data, aes(x = bmi, y =  
water)) +
```

```
  geom_point() +
```

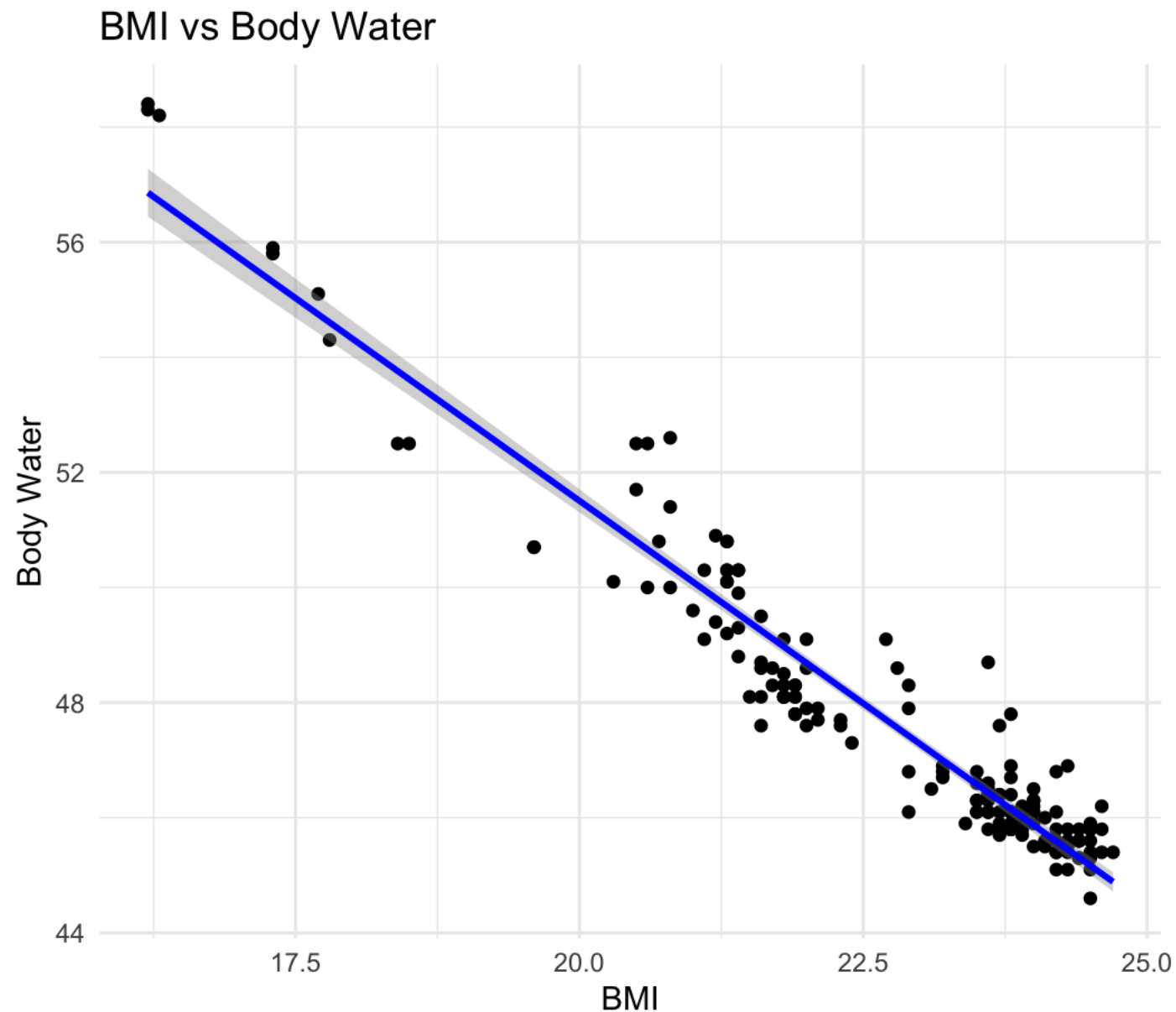
```
  geom_smooth(method = "lm",  
color = "blue") +
```

```
  labs(title = "BMI vs Body  
Water",
```

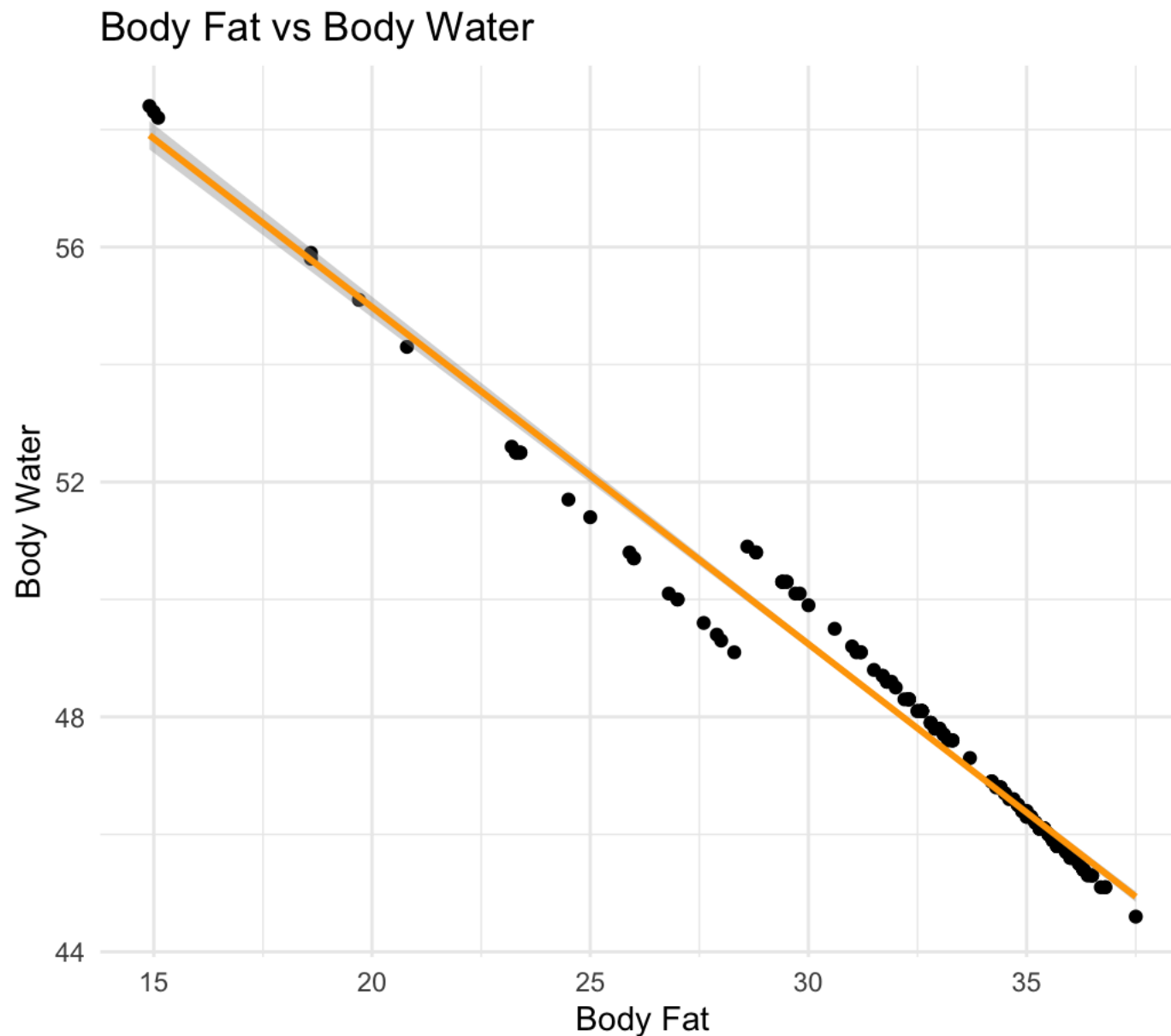
```
        x = "BMI",
```

```
        y = "Body Water") +
```

```
  theme_minimal()
```



```
ggplot(data, aes(x = fat, y =  
water)) +  
  geom_point() +  
  geom_smooth(method = "lm", color  
= "blue") +  
  labs(title = "Body Fat vs Body  
Water",  
        x = "Body Fat",  
        y = "Body Water") +  
  theme_minimal()
```



```
cor.test(data$bmi, data$fat)
```

Pearson's product-moment correlation

data: data\$bmi and data\$fat

$t = 42.274$ ,  $df = 166$ , **p-value < 2.2e-16**

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

0.9415115 0.9678006

sample estimates:

cor

**0.9565598**

A very strong positive correlation was observed between **BMI** and **body fat** ( $r = 0.96$ ,  $p < 0.001$ ).

```
cor.test(data$bmi, data$water)
```

Pearson's product-moment correlation

data: data\$bmi and data\$water

$t = -45.006$ ,  $df = 166$ , **p-value < 2.2e-16**

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

-0.9713931 -0.9479596

sample estimates:

cor

**-0.9613819**

A very strong negative correlation was observed between **BMI** and **body water** ( $r = -0.96$ ,  $p < 0.001$ ).

```
cor.test(data$fat, data$water)
```

Pearson's product-moment correlation

data: data\$fat and data\$water

$t = -85.595$ ,  $df = 166$ , **p-value < 2.2e-16**

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

-0.9917779 -0.9849148

sample estimates:

cor

**-0.9888601**

A very strong negative correlation was observed between **body fat** and **body water** ( $r = -0.99$ ,  $p < 0.001$ ).